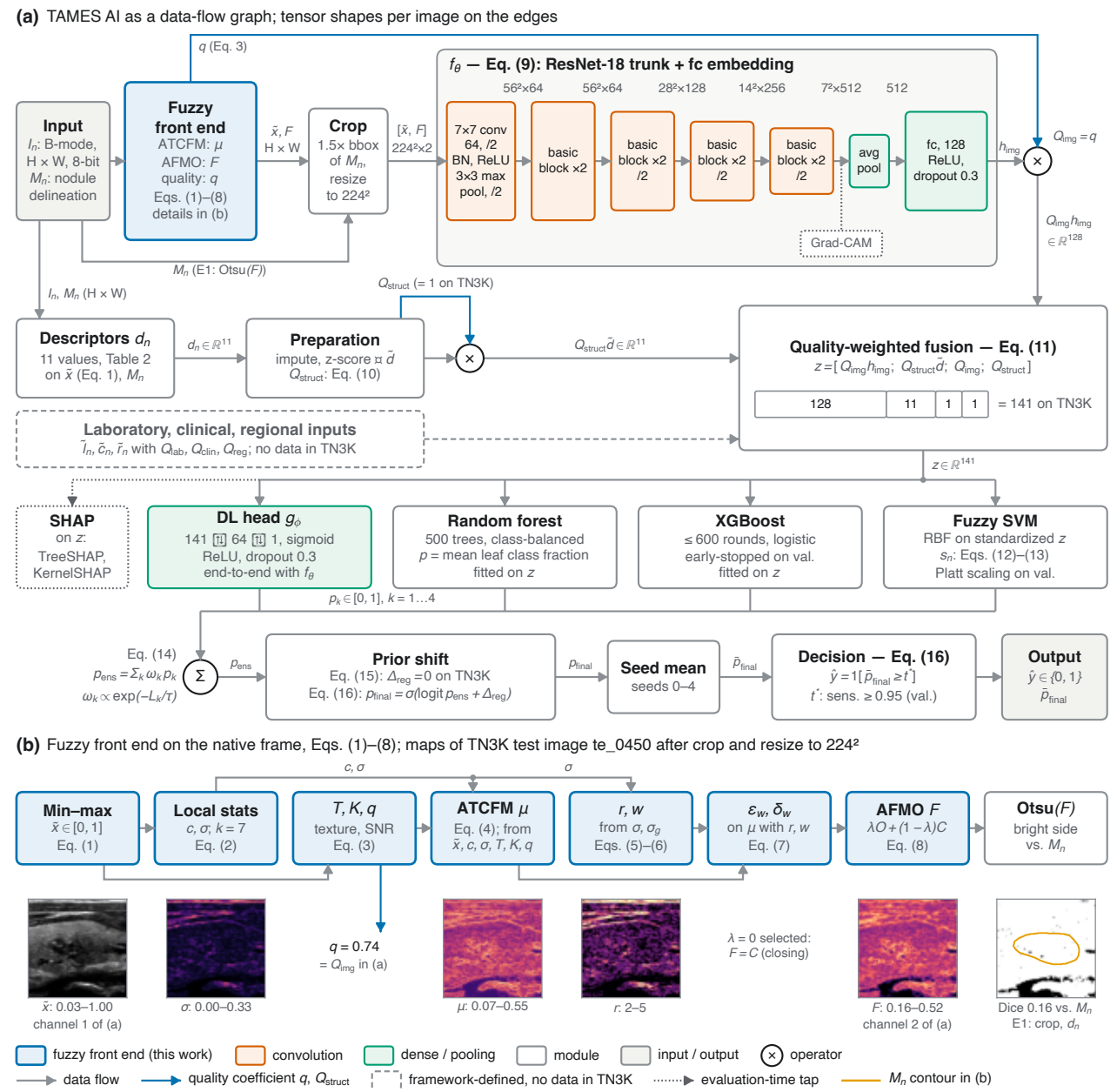

(a) TAMES AI as a data-flow graph; tensor shapes per image on the edges
q (Eq. 3)
Input
I_n: B-mode, H × W, 8-bit
M_n: nodule delineation
Fuzzy front end
ATCFM: μ
AFMO: F
quality: q
Eqs. (1)–(8)
details in (b)
$\tilde{x}, F$
H × W
Crop
1.5× bbox of M_n, resize to 224²
$[\tilde{x}, F]$
224²×2
f_θ — Eq. (9): ResNet-18 trunk + fc embedding
56²×64
56²×64
28²×128
14²×256
7²×512
512
7×7 conv 64, /2 BN, ReLU 3×3 max pool, /2
basic block ×2
basic block ×2 /2
basic block ×2 /2
basic block ×2 /2
avg pool
fc, 128 ReLU, dropout 0.3
h_{img}
$Q_{img} = q$
$Q_{img} h_{img}$ $\in \mathbb{R}^{128}$
Grad-CAM
M_n (E1: Otsu(F))
I_n, M_n (H × W)
Q_{struct} (= 1 on TN3K)
Descriptors d_n
11 values, Table 2 on $\tilde{x}$ (Eq. 1), M_n
$d_n \in \mathbb{R}^{11}$
Preparation
impute, z-score ¤ $\tilde{d}$
Q_{struct}: Eq. (10)
$Q_{struct} \tilde{d} \in \mathbb{R}^{11}$
Quality-weighted fusion — Eq. (11)
$z = [Q_{img} h_{img};\ Q_{struct} \tilde{d};\ Q_{img};\ Q_{struct}]$
128
11
1
1
= 141 on TN3K
Laboratory, clinical, regional inputs
$\tilde{l}_n, \tilde{c}_n, \tilde{r}_n$ with $Q_{lab}, Q_{clin}, Q_{reg}$; no data in TN3K
$z \in \mathbb{R}^{141}$
SHAP
on z:
TreeSHAP,
KernelSHAP
DL head g_ϕ
141 ⇅ 64 ⇅ 1, sigmoid
ReLU, dropout 0.3
end-to-end with f_θ
Random forest
500 trees, class-balanced
p = mean leaf class fraction
fitted on z
XGBoost
≤ 600 rounds, logistic
early-stopped on val.
fitted on z
Fuzzy SVM
RBF on standardized z
s_n: Eqs. (12)–(13)
Platt scaling on val.
$p_k \in [0, 1]$, $k = 1 \ldots 4$
Eq. (14)
$p_{ens} = \Sigma_k \omega_k p_k$
$\omega_k \propto \exp(-L_k/\tau)$
Σ
p_{ens}
Prior shift
Eq. (15): $\Delta_{reg} = 0$ on TN3K
Eq. (16): $p_{final} = \sigma(\text{logit}\, p_{ens} + \Delta_{reg})$
p_{final}
Seed mean
seeds 0–4
$\bar{p}_{final}$
Decision — Eq. (16)
$\hat{y} = 1[\bar{p}_{final} \geq t^*]$
t^*: sens. ≥ 0.95 (val.)
Output
$\hat{y} \in \{0, 1\}$
$\bar{p}_{final}$
(b) Fuzzy front end on the native frame, Eqs. (1)–(8); maps of TN3K test image te_0450 after crop and resize to 224²
c, σ
σ
Min–max
$\tilde{x} \in [0, 1]$
Eq. (1)
Local stats
c, σ; $k = 7$
Eq. (2)
T, K, q
texture, SNR
Eq. (3)
ATCFM μ
Eq. (4); from $\tilde{x}, c, \sigma, T, K, q$
r, w
from σ, σ_g
Eqs. (5)–(6)
ε_w, δ_w
on μ with r, w
Eq. (7)
AFMO F
$\lambda O + (1 - \lambda) C$
Eq. (8)
Otsu(F)
bright side vs. M_n
$\tilde{x}$: 0.03–1.00
channel 1 of (a)
σ: 0.00–0.33
$q = 0.74$
= Q_{img} in (a)
μ: 0.07–0.55
r: 2–5
$\lambda = 0$ selected:
$F = C$ (closing)
F: 0.16–0.52
channel 2 of (a)
Dice 0.16 vs. M_n
E1: crop, d_n
fuzzy front end (this work)
convolution
dense / pooling
module
input / output
operator
data flow
quality coefficient q, Q_{struct}
framework-defined, no data in TN3K
evaluation-time tap
M_n contour in (b)